
REQUIREMENTS NOT MET

- N/A

VIDEO FILE LINK

https://youtube.com/shorts/t7zGID5FI_g?si=RPE3gLnGu-ci8uR2

PROBLEMS ENCOUNTERED

N/A

FUTURE WORK/APPLICATIONS

A traffic light controller by itself is useful, and can be used to safely allow emergency vehicles through intersections immediately 😊, we also use VHDL for the first time, which is MUCH faster to design in than a BDF overall. This will allow us to develop counters and more complex logic faster than before.

[illegible]

D LOGIC

$$D3 = (\text{not } Q3 * Q2 * Q1 * Q0) + (Q3 * \text{not } Q2 * \text{not } Q1 * \text{not } Q0) + (EV * Q3 * \text{not } Q2 * \text{not } Q1)$$

$$D2 = AC'D' + AC'E' + C'D'EF + C'DE'F + C'DEF' + BC'DE'$$

$$= (EV * \text{not } Q3 * \text{not } Q2) + (EV * \text{not } Q3 * \text{not } Q1) + (\text{not } Q3 * \text{not } Q2 * Q1 * Q0) + (\text{not } Q3 * Q2 * Q1 * \text{not } Q0) + (CW * \text{not } Q3 * Q2 * \text{not } Q1)$$

$$D1 = C'E'F + C'EF' + AC'D' + AC'E'$$

$$= (\text{not } Q3 * \text{not } Q1 * Q0) + (\text{not } Q3 * Q1 * \text{not } Q0) + (EV * \text{not } Q3 * \text{not } Q2) + (EV * \text{not } Q3 * \text{not } Q1)$$

$$D0 = A'C'D'F' + A'D'E'F' + C'DEF' + A'BC'F' + ACD'E'$$

$$= (\text{not } EV * \text{not } Q3 * \text{not } Q2 * \text{not } Q0) + (\text{not } EV * \text{not } Q2 * \text{not } Q1 * \text{not } Q0) + (\text{not } Q3 * Q2 * Q1 * \text{not } Q0) + (\text{not } EV * CW * \text{not } Q3 * \text{not } Q0) + (EV * Q3 * \text{not } Q2 * \text{not } Q0)$$

Lab 5, Smart Traffic Light Controller

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Class #: 11081

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Description: Develop a smart traffic light controller utilizing VHDL for customized components.

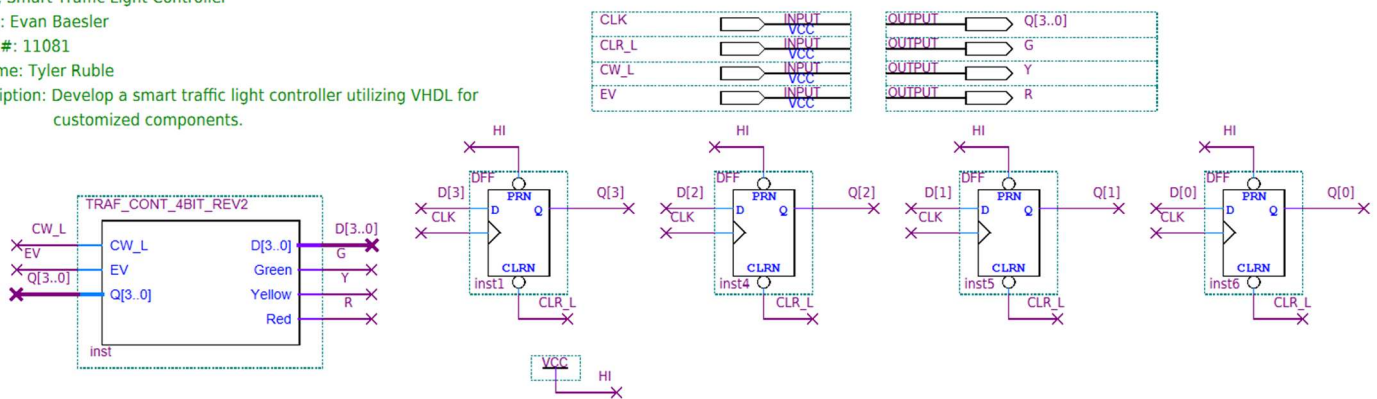


Figure 2: Quartus BDF for Traffic State Controller

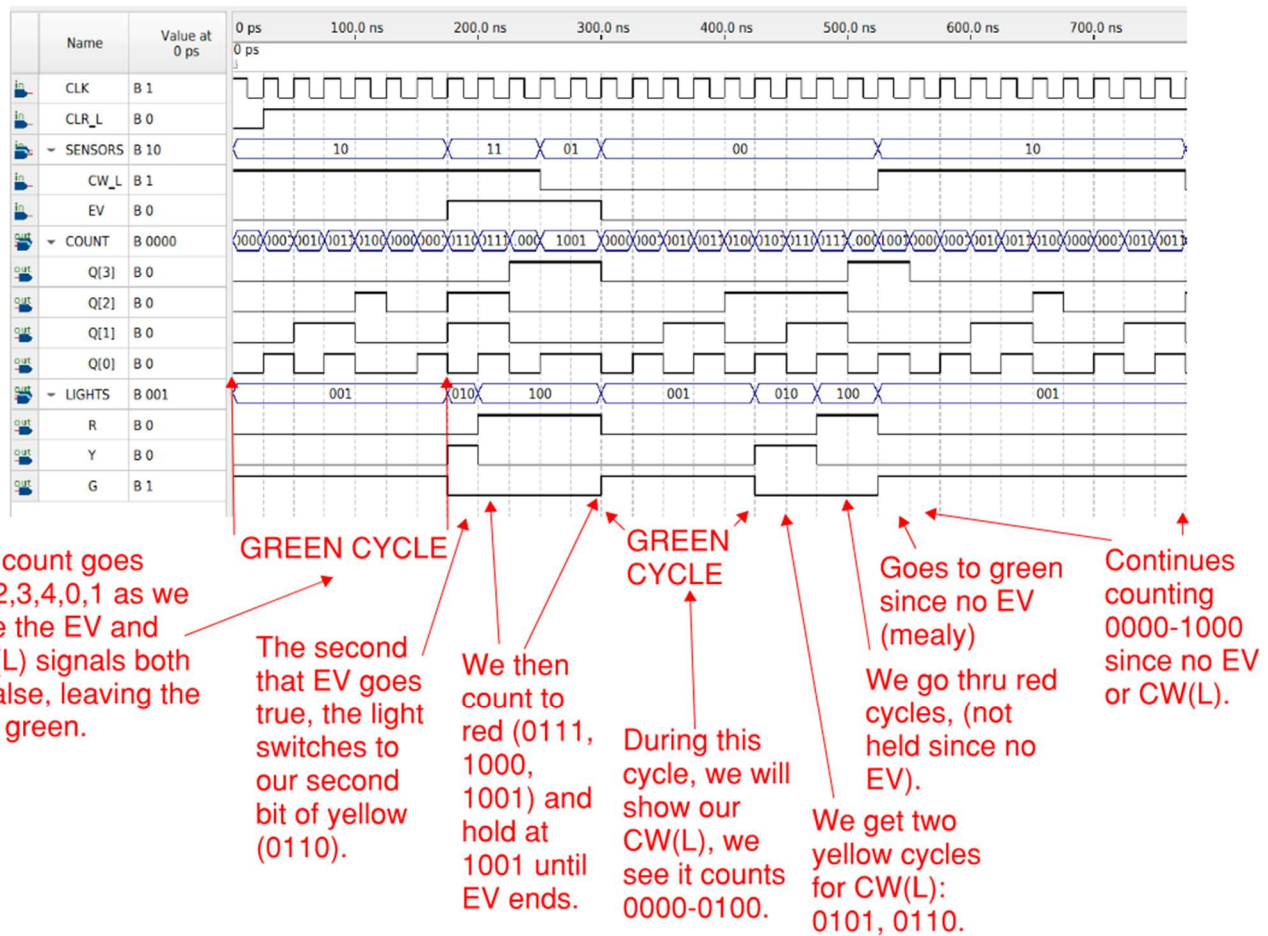


Figure 3: Annotated Simulation of Traffic Controller

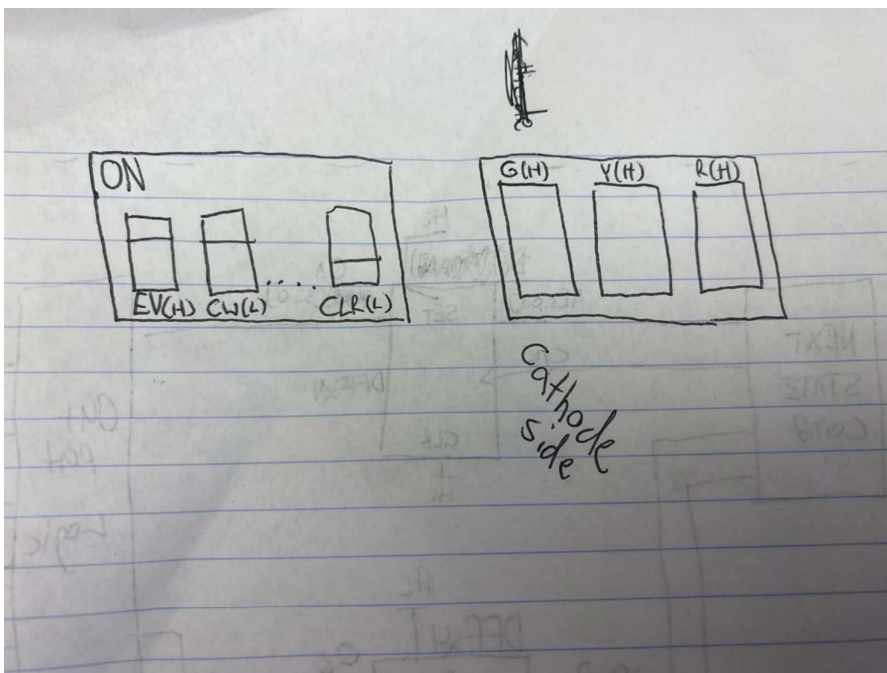


Figure 4: Switch and LED Diagrams